

CS 59300 (Application of Deep Learning) (Fall 2025)

Show how to create a VM in **Google Cloud**, download the code from Hugging Face Space, install python packages, and run the Gradio App website at the Google Cloud.

1. Request a Google Cloud coupon by accessing the URL posted in the course Announcements section of Brightspace. Please use your school email (i.e., pfw.edu email).
2. Login into Google Cloud at <https://cloud.google.com/>. Create a new project “CS593 App of DL”, and link it to “Billing Account for Education” billing account.
3. In the project, please enable “Compute Engine API” and create a VM instance with “e2-medium” machine type, like

Machine configuration

Name *
instance-20250911-174447

Region *
us-central1 (Iowa)

Zone *
Any

Region is permanent

Google will choose a zone on your behalf, maximizing VM obtainability. Zone is permanent.

☒ General purpose ☐ Compute optimized ☐ Memory optimized ☐ Storage optimized ☐ GPUs

Machine types for common workloads, optimized for cost and flexibility

	Series	Description	vCPUs	Memory	CPU Platform
☐	C4	Consistently high performance	2 - 288	4 - 2,232 GB	Intel Emerald
☐	C4A	Arm-based consistently high performance	1 - 72	2 - 576 GB	Google Axion
☐	C4D	Consistently high performance	2 - 384	3 - 3,072 GB	AMD Turin
☐	N4	Flexible & cost-optimized	2 - 80	4 - 640 GB	Intel Emerald
☐	C3	Consistently high performance	4 - 192	8 - 1,536 GB	Intel Sapphire
☐	C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMD Genoa
☒	E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Intel Broadwell
☐	N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel Cascade
☐	N2D	Balanced price & performance	2 - 224	2 - 896 GB	AMD Milan
☐	T2A	Scale-out workloads	1 - 48	4 - 192 GB	Ampere Altra
☐	T2D	Scale-out workloads	1 - 60	4 - 240 GB	AMD Milan
☐	N1	Balanced price & performance	0.25 - 96	0.6 - 624 GB	Intel Haswell

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. [Learn more](#)

☒ Preset ☐ Custom

e2-medium (2 vCPU, 1 core, 4 GB memory)

[Equivalent code](#)

Monthly estimate

\$34.46

That's about \$0.05 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$24.46
100 GB balanced persistent disk	\$10.00
Logging	Cost varies
Monitoring	Cost varies
Snapshot schedule	Cost varies
Total	\$34.46

[Compute Engine pricing](#)

[Cloud Operations pricing](#)

[Less](#)

Please select 100 GB for the disk in the “OS and storage” tab.

Please check “Allow HTTP traffic” and “Allow HTTPS traffic” for “Firewall” in the “Networking” tab.

4. Login into VM at Google Cloud by clicking “Open in browser window” under “SSH” tab.

Install python and Git, as well as related packages, by running:

```
$ sudo apt update
```

```
$ sudo apt install python3 python3-dev python3-venv python3-pip
```

```
$ python3 --version
```

```
$ pip --version
```

```
$ sudo apt install git
```

```
$ sudo apt install ffmpeg
```

5. Download the code form your Hugging Face Space for Whisper application (**Note: should be your own project at Hugging Face Space**):

```
$ git clone https://huggingface.co/spaces/zeshengchen/test_whisper
```

6. Run the following commands:

```
$ cd test_whisper/
```

```
$ python3 -m venv env
```

```
$ source env/bin/activate
```

```
$ pip install -r requirements.txt
```

```
$ pip install gradio
```

```
$ nohup python3 -u app.py > output.log &
```

7. After 1 or 2 minutes, run “cat output.log” command and then go to the website shown in the output, like “https://ad57c6b8b267c1c8f0.gradio.live”.
8. To stop the application, run “ps aux | grep python3” command to find the process id and then run a command like “kill -9 9260” to kill the process.
9. Use “deactivate” command to remove the environment for running this application. Next time to run it again, use only “source env/bin/activate” and “nohup python3 -u app.py > output.log &” commands.